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Manual of Neonatal Respiratory Care



Fourth Edition

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Neonatal Pulmonary Graphics

22

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I. Indications
   A. Optimizing mechanical ventilation parameters
      1. Peak inspiratory pressure (PIP)
      2. Positive end expiratory pressure (PEEP)
      3. Inspiratory and expiratory tidal volume (V_{TI} or V_{TE})
      4. Inspiratory time (T_I)
      5. Expiratory time (T_E)
      6. Flow rate
      7. Synchronization
      8. Compliance
   B. Evaluation of infant's spontaneous effort
      1. Spontaneous V_T
      2. Minute ventilation (MV)
      3. Respiratory pattern
      4. Readiness for extubation
   C. Therapeutic response to pharmacologic agents
      1. Surfactant
      2. Bronchodilators
      3. Diuretics
      4. Steroids

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D. Evaluation of respiratory waveforms, loops, and mechanics
   1. Waveforms
      a. Pressure
      b. Flow
      c. Volume
   2. Loops
      a. Pressure–volume loop
      b. Flow–volume loop
   3. Mechanics
      a. Dynamic compliance (C_D) or static compliance (C_{ST})
      b. Resistance (inspiratory and expiratory)
      c. Time constants

E. Disease Evaluation
   1. Restrictive
   2. Obstructive
   3. Severity
   4. Recovery

II. Graphical User Interfaces

A. Graphical user interfaces (GUI) provide continuous, real-time, breath-to-breath feedback of the interaction between the patient and the ventilator.

B. They are also an excellent teaching tool.

C. Graphics monitors have been available for the last decade as an option that can be added to ventilators; now, the latest generation of ventilators has touch screen interfaces with color displays that are integral to the ventilator.

D. Graphics data collection. Flow sensor location:
   1. Proximal flow sensor positioning, at the airway opening, is critical for accurate waveforms, loops, and data.
   2. Distal flow sensors, within the ventilator or near the tubing exit, produce waveforms, loops, and data that include circuit compliance and resistance. So-called circuit compliance compensation calculations do not accurately correct the data displayed. They should not be used in very preterm babies, where suboptimal volume delivery may promote lung injury.

E. Flow sensors
   1. Heated wire anemometer: Measures the amount of current required to keep a heated wire at a constant temperature as gas flows past the wire and heat is convected. This current can be converted to a flow measurement, and integrated to determine volume.
   2. Differential pressure pneumotachometer: As gas flows through the sensor across an element, a differential pressure is created between the upstream and downstream sensing ports. The change in pressure across the element is proportional to flow.
   3. Diaphragmatic neural sensor: This technique uses a modified feeding tube containing a number of electrode sensors for measurement of diaphragmatic EMG.

F. Neonatal-capable ventilators with integral GUI
   1. Avea (CareFusion, Yorba Linda, CA)
   2. Dräger Babylog VN500, Evita XL, Evita Infinity V500 (Draeger Medical, Inc, Telford, PA)
   3. Puritan Bennett 840 (Covidien-Puritan Bennett, Mansfield, MA)
   4. Servo-i (Maquet Critical Care, Wayne, NJ)
   5. Hamilton C-2 and S-1 (Hamilton Medical, Reno, NV)
   6. SLE 4000 and 5000 (SLE, Ltd., Surrey, UK)

7. Newport e360T and Newport WAVE (if Compass added) (Newport Medical Instruments, Newport Beach, CA)

G. Neonatal-pediatric ventilators that are still in use, but not currently being manufactured
1. VIP BIRD/GOLD with Bird Graphic Monitor (CareFusion Health Care, Yorba Linda, CA)
2. Bear Cub 750 with Ventilator Graphics Monitor (CareFusion Healthcare, Yorba Linda, CA).
3. Dräger Babylog 8000+ (Dräger, Telford, PA)

III. Graphic Waveforms

A. Pressure
1. Pressure waveform (Fig. 22.1, top waveform)
   a. The up-sweep of the waveform represents inspiration and the down-sweep represents expiration.
   b. PIP is the maximum pressure point on the curve (A).
   c. PEEP is the baseline pressure level (B).
   d. The area under the curve represents the mean airway pressure (shaded—Fig. 22.2).
   e. The shape of the curve represents the breath type, e.g., volume (triangular) or pressure (square).

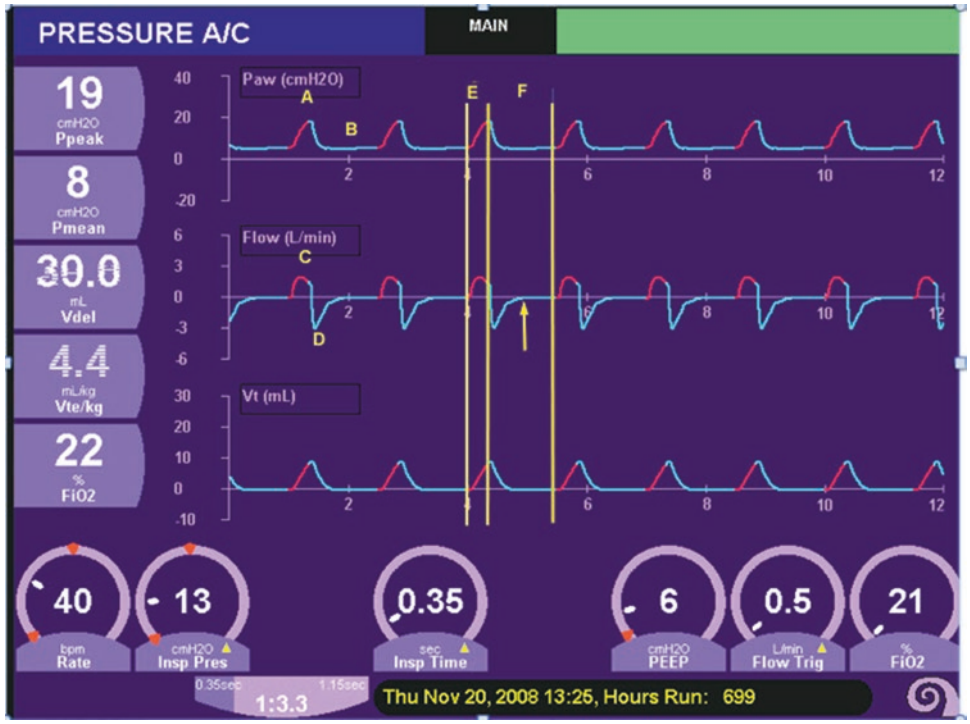


Fig. 22.1 The pressure waveform. See text for full description

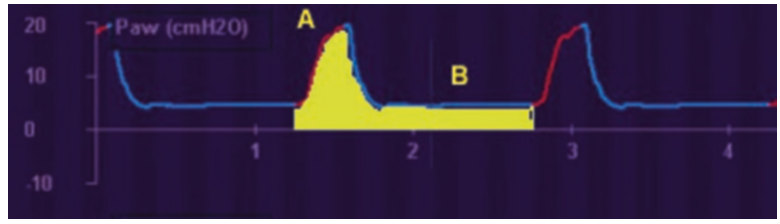


Fig. 22.2 Graphic display of mean airway pressure



Fig. 22.3 Scalar tracing showing the effect of prolonging inspiratory time. Panel (**a**) shows inspiratory time set such that inspiration ends when flow returns to zero. Panel (**b**) shows a prolonged inspiratory time, with a pressure plateau in which no further volume delivery occurs. Mean airway pressure is increased

B. Flow
   1. Flow waveform (Fig. 22.1, center waveform)
      a. Horizontal line is the zero flow point. Up-sweep of the flow waveform above this line is inspiratory flow, and down-sweep is expiratory flow.
      b. Greatest deflection above reference equals peak inspiratory flow (C).
      c. Greatest deflection below reference equals peak expiratory flow (D).
      d. Inspiratory time is measured from the initial flow delivery until expiratory flow begins (E).
      e. At the point on the waveform where flow is zero (Fig. 22.3, arrows), no additional volume can be delivered to the infant. Panel a shows inspiratory time set such that inspiration ends when flow returns to zero. Panel b shows a prolonged inspiratory time, with a pressure plateau in which no further volume delivery occurs. Mean airway pressure is increased.
      f. Flow cycling allows a mechanical breath to be triggered (cycled) into expiration by a specific algorithm (usually 5–25 % of peak inspiratory flow). The ability of a patient to control inspiratory time and cycle a breath to expiration may lead to improved synchronization. This feature is available on the newer generation ventilators and on any ventilator having pressure support (Fig. 22.4; red shows inspiration, blue expiration; arrow shows end-inspiration above zero flow baseline).
      g. Expiratory time is the point where expiratory flow begins until the next inspiration begins (Fig. 22.1F). When expiratory flow returns to zero, lung deflation is complete (Fig. 22.1, arrow).
      h. If flow has not reached zero before the next breath is delivered, gas trapping may occur (Fig. 22.5, arrow).

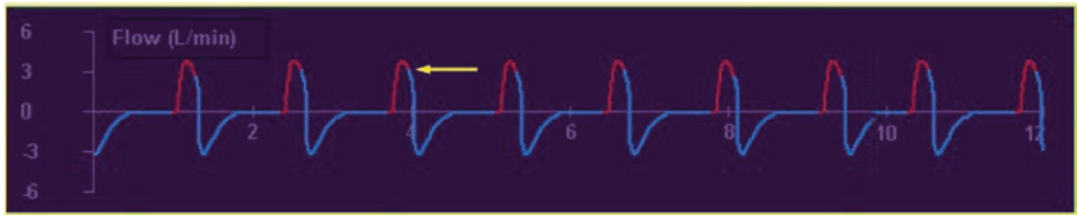


Fig. 22.4 This flow waveform illustrates flow cycling. *Red* shows inspiration, *blue* expiration. *Arrow* shows end inspiration above zero flow baseline

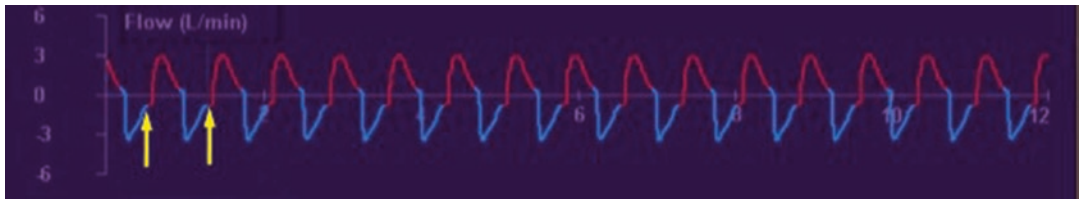


Fig. 22.5 Flow waveform demonstrating gas trapping

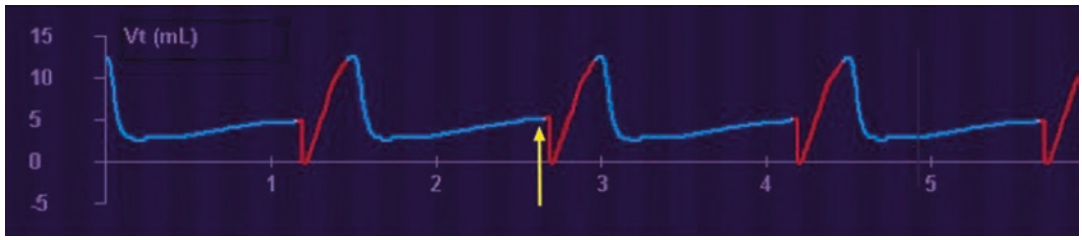


Fig. 22.6 Flow waveform demonstrating an endotracheal tube leak. Note that expiratory flow does not completely return to baseline before the next breath

C. Volume
   1. Volume waveform (Fig. 22.1, bottom waveform)
      a. Inspiration is represented as the waveform sweeps upward and expiration as the waveform sweeps downward.
      b. The red line represents delivered inspiratory tidal volume.
      c. An endotracheal tube leak is observed when the expiratory portion of the waveform fails to return to the zero baseline (Fig. 22.6, arrow).
   2. Traditional volume ventilation produces a square-flow waveform. Some ventilators enable this to be decelerated.

D. Patient-ventilator interaction
   1. Intermittent mandatory ventilation (IMV)
      a. Unsynchronized IMV, the initial form of neonatal ventilation that allowed patients to breath between ventilator cycles, results in machine breaths delivered at various times in the patient effort cycle with deleterious results (Fig. 22.7). Complications include pneumothorax and IVH. Yellow arrows show patient effort augmenting the ventilator breath, with very high Vt. Green arrows show patient beginning exhalation during machine inspiration.
      b. Synchronized IMV (SIMV) synchronizes patient inspiratory effort to mechanical breath delivery (Fig. 22.8). Yellow arrows show a synchronized machine breath, and red arrows show an unaugmented spontaneous breath.

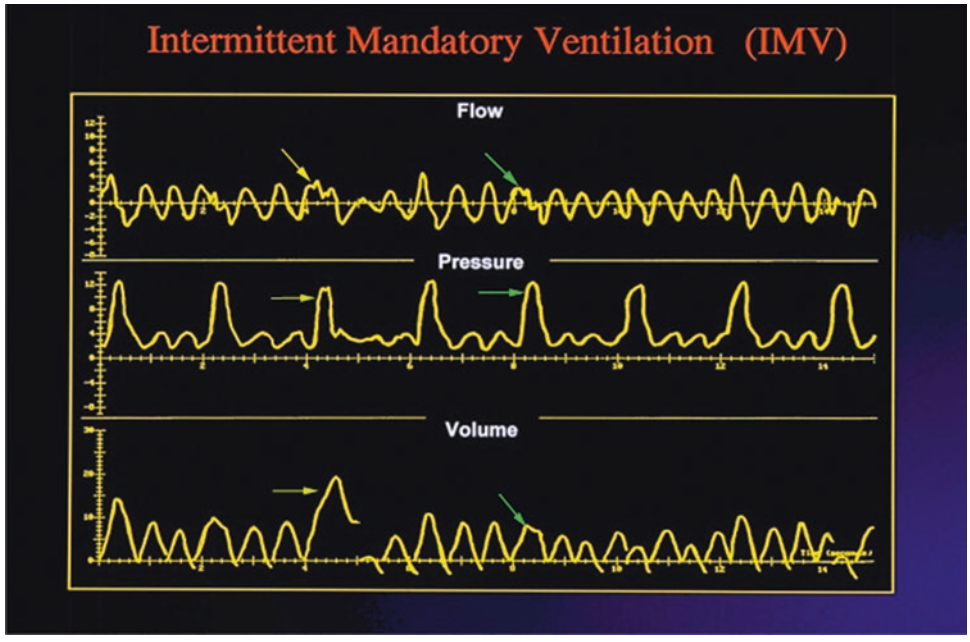


Fig. 22.7 Intermittent mandatory ventilation. *Yellow arrows* show patient effort augmenting the ventilator breath, with very high Vt. *Green arrows* show patient beginning exhalation during machine inspiration

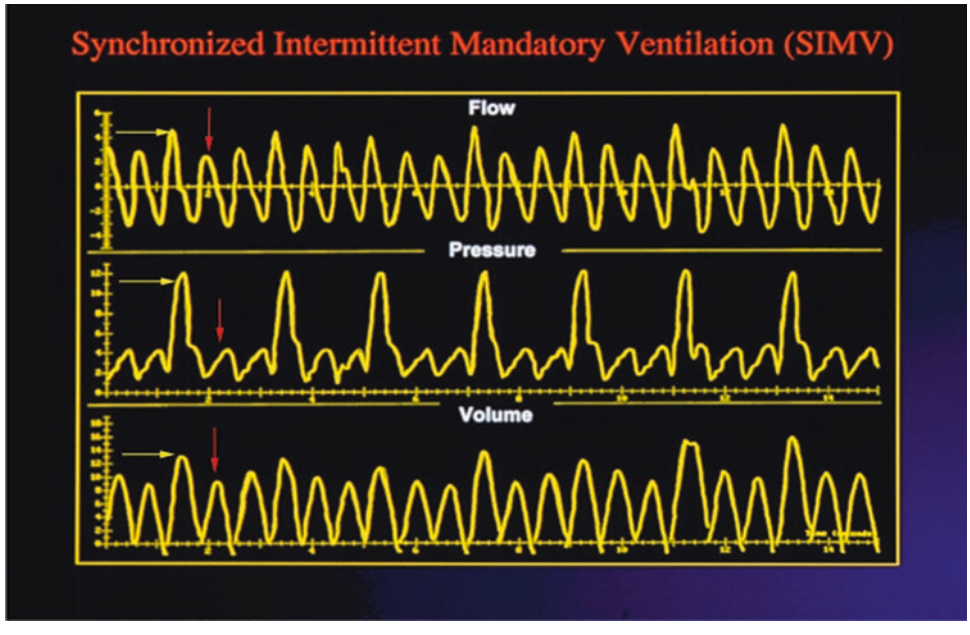


Fig. 22.8 Synchronized IMV. *Yellow arrows* show a synchronized machine breath, *red arrows* show an unaugmented spontaneous breath

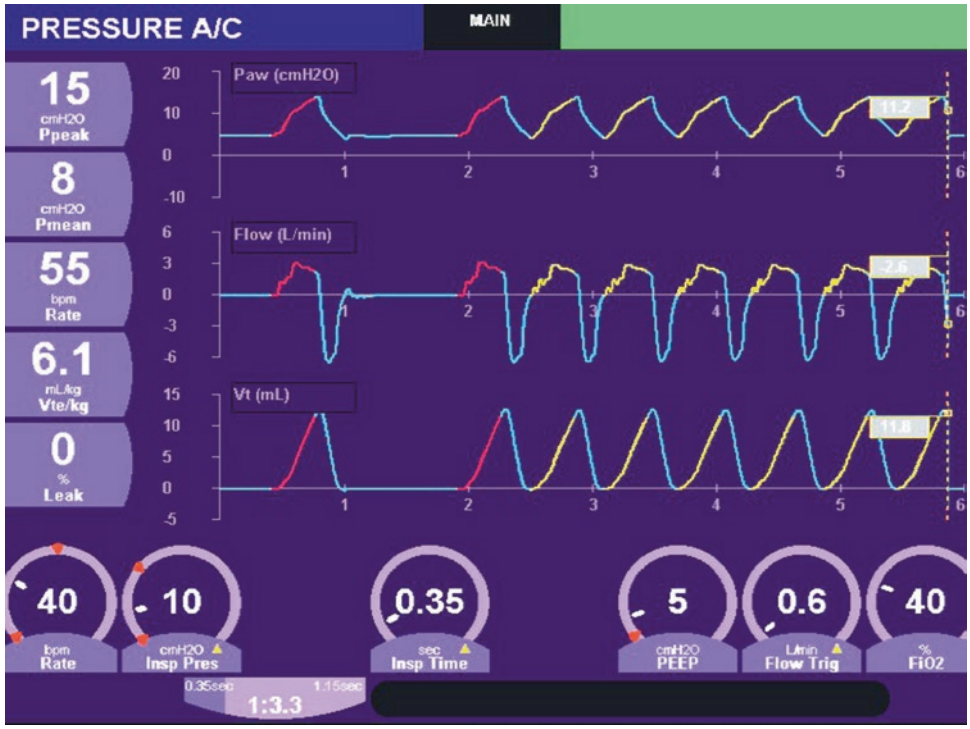


Fig. 22.9 Assist/control. *Red* shows a time-cycled machine delivered breath during apnea, *yellow* shows patient-initiated breaths

2. Assist/control and pressure support
   a. Assist/control (A/C) assists (triggers a machine-delivered inspiration) when the patient initiates a breath, and controls (delivers a time-cycled inspiration) if the patient is apneic or fails to trigger (Fig. 22.9). Red shows a time-cycled machine delivered breath during apnea, and yellow shows patient initiated breaths.
   b. Pressure support is very similar to A/C as all patient-triggered breaths are supported by a set pressure. Unlike A/C, it may be used as a blended mode during SIMV (Fig. 22.10).

IV. Graphic Loops
   A. Pressure–volume (*P-V*) loop (Fig. 22.11)
      1. A pressure–volume loop displays the relationship of pressure to volume.
         a. Pressure is displayed along the horizontal axis and volume is displayed on the vertical axis.
         b. Inspiration is represented by the up-sweep from the baseline (PEEP) terminating at PIP. Expiration is the down-sweep from PIP back to baseline.
         c. A line drawn from each endpoint represents pulmonary compliance ($\Delta V/\Delta P$).
         d. The *P-V* loop may be used to assess adequacy of PEEP, used to maintain end-expiratory lung volume (Fig. 22.12). If the inspiratory limb of the *P-V* curve demonstrates a lower inflection point, identifying opening pressure, PEEP is inadequate.
         e. The *P-V* loop may help identify lung overdistension (Fig. 22.13). If the inspiratory limb flattens at the top, this indicates pressure exposure without further volume delivery. It is measured on most neonatal ventilators as the C_{20}/C ratio.

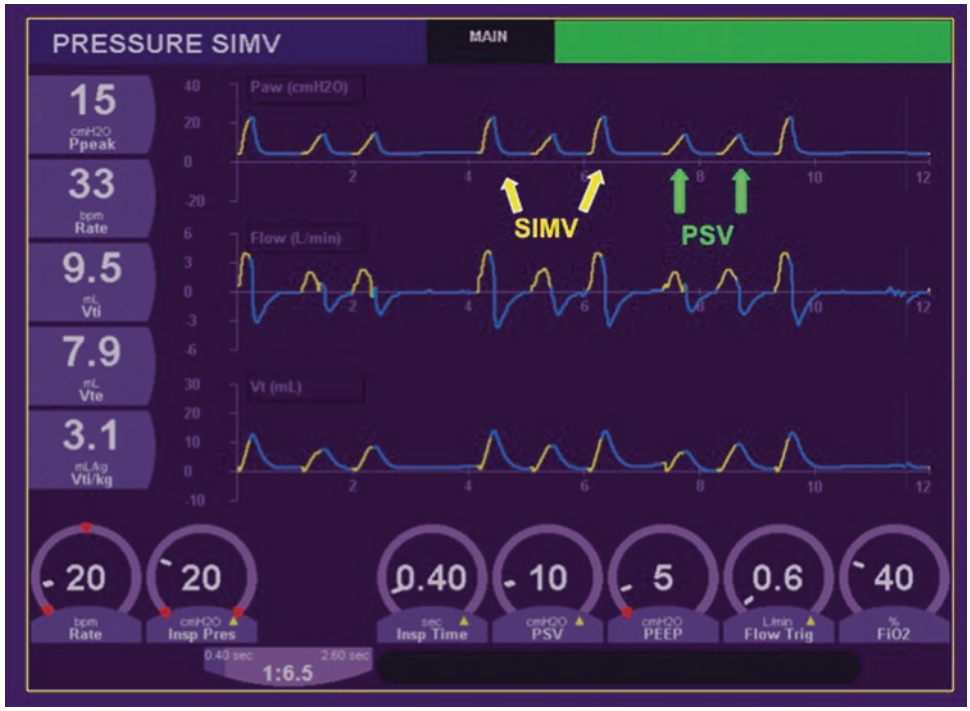


Fig. 22.10 Pressure support during SIMV

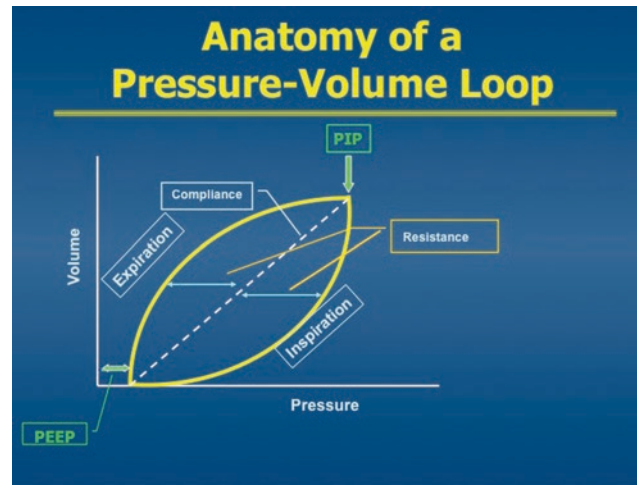


Fig. 22.11 The pressure –volume loop

f. *P-V* loops can help evaluate whether flow delivery from the ventilator is adequate to meet the needs of the patient. Inadequate flow is represented by cusping of the inspiratory portion of the curve. Severe flow limitation may appear as a "figure-8" on the *P-V* loop (Fig. 22.14).

B. Flow-Volume ($\dot{V} - V$) loop (Fig. 22.15)

1. A $\dot{V} - V$ loop displays the relationship between volume and flow. Volume is plotted on the horizontal axis and flow is plotted on the vertical axis. The breath starts at the zero axis

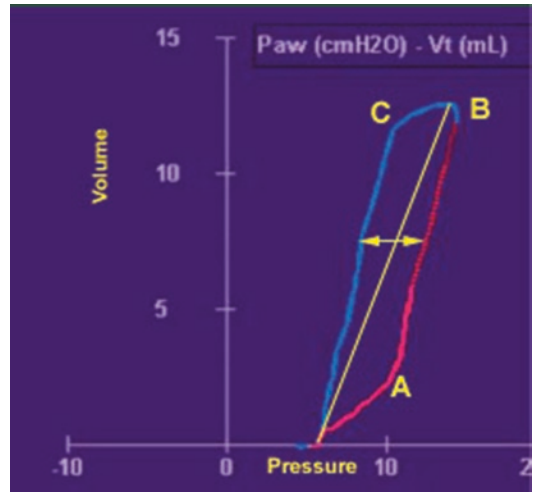


Fig. 22.12 PEEP assessment using the *P-V* loop

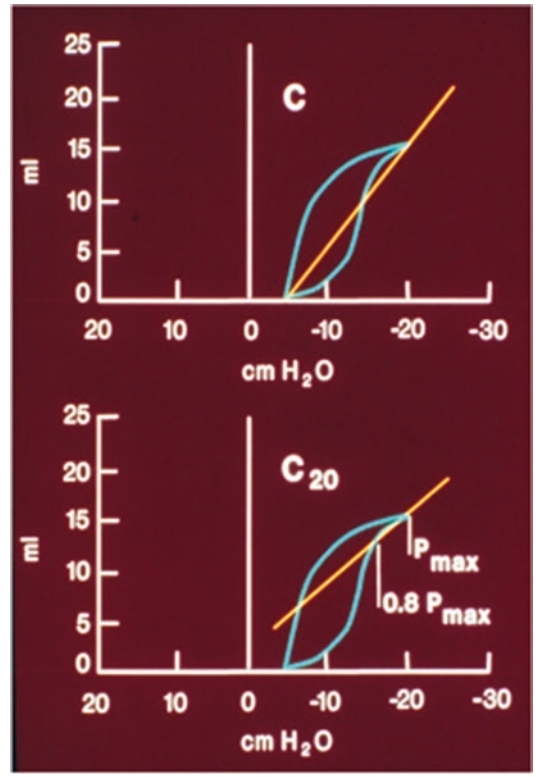


Fig. 22.13 Lung overdistension as assessed by the C_{20}/C ratio

and moves upward and to the right on inspiration, terminating at the delivered inspiratory volume and downward, to the left, back to zero on expiration.

a. The $\dot{V} - V$ loop changes shape when either inspiratory resistance (Fig. 22.16, with flattened inspiratory limb) or expiratory resistance (Fig. 22.17, with flattened expiratory limb) is increased.

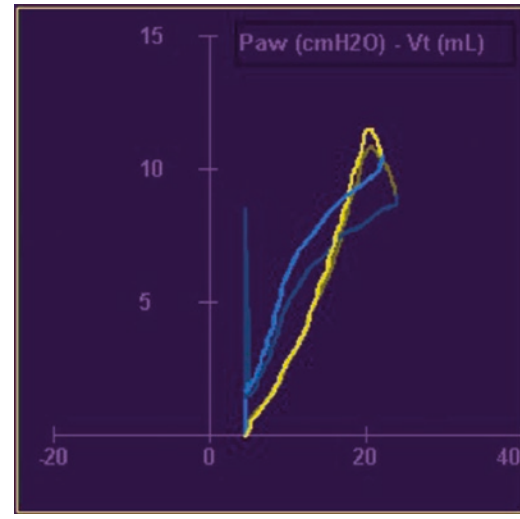


Fig. 22.14 *P-V* loop showing inadequate inspiratory flow resulting in figure 8 appearance

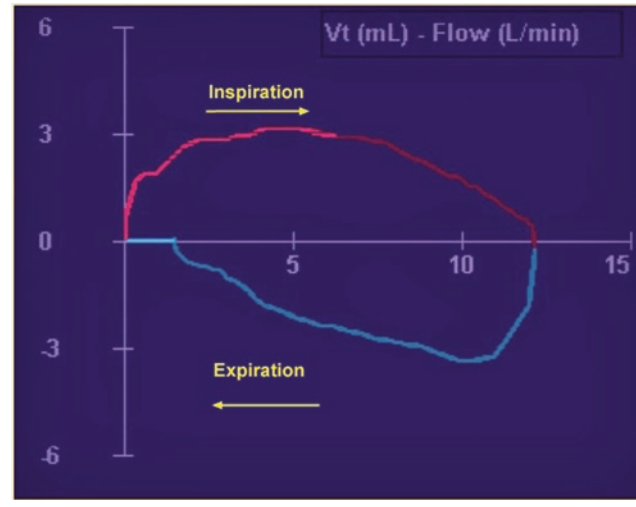


Fig. 22.15 The flow -volume loop

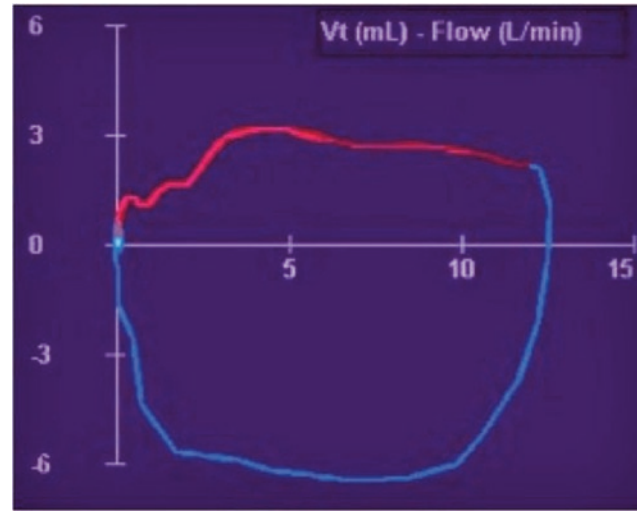


Fig. 22.16 $\dot{V} - V$ loop showing increased inspiratory resistance

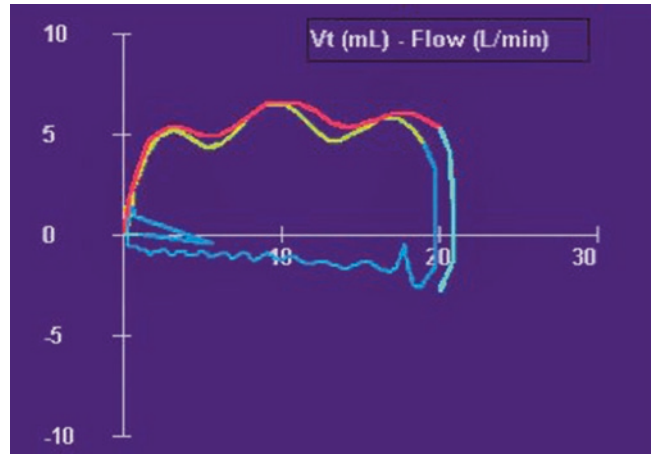


Fig. 22.17 $\dot{V} - V$ loop showing increased expiratory resistance

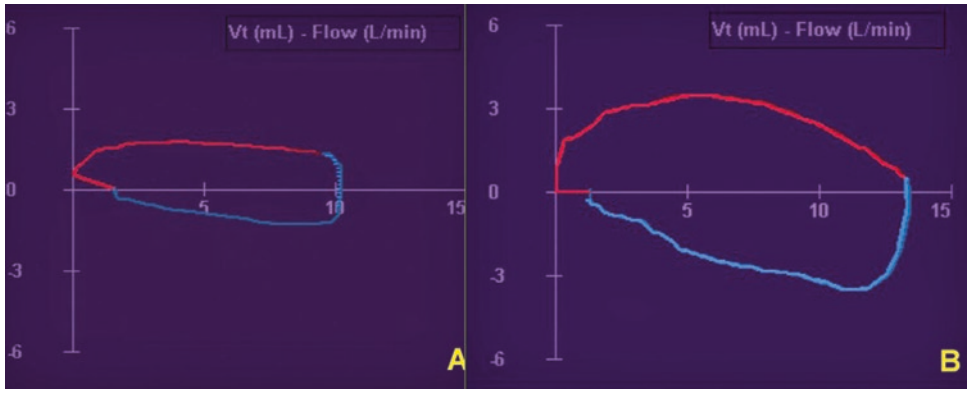


Fig. 22.18 $\dot{V} - V$ loop showing bronchodilator effect. (**a**) Pretreatment. (**b**) Posttreatment

b. The $\dot{V} - V$ loop is useful for evaluating the effectiveness of bronchodilators in treating airway reactivity. In Fig. 22.18, increased inspiratory and expiratory flow is seen in loop B as compared to the loop A.
c. Presence of secretions or water in the ventilator tubing or flow sensor can be seen on the loop displays. Since suctioning should only be performed as indicated, loops are a useful way to evaluate the need for suctioning or draining water from the circuit (Fig. 22.19).

V. Dynamics Measurements/Calculations
A. Tidal volume is measured on inspiration and expiration. Normal delivered V_T is 4–7 mL/kg.
B. Minute ventilation is the product of V_T and respiratory rate. The normal range is 240–360 mL/kg/min.
C. Pressure may be measured as peak inspiratory pressure or static pressure. Static pressure is obtained by doing an inflation hold maneuver, which measures pressure obtained by closing the exhalation valve and stopping flow delivery during a mechanical breath.
D. Compliance is the relationship between a change in volume and a change in pressure.
1. Dynamic compliance (C_D) is the measurement of compliance based on peak pressure:

$$C_D = \frac{V_{Ti}}{PIP - PEEP}$$

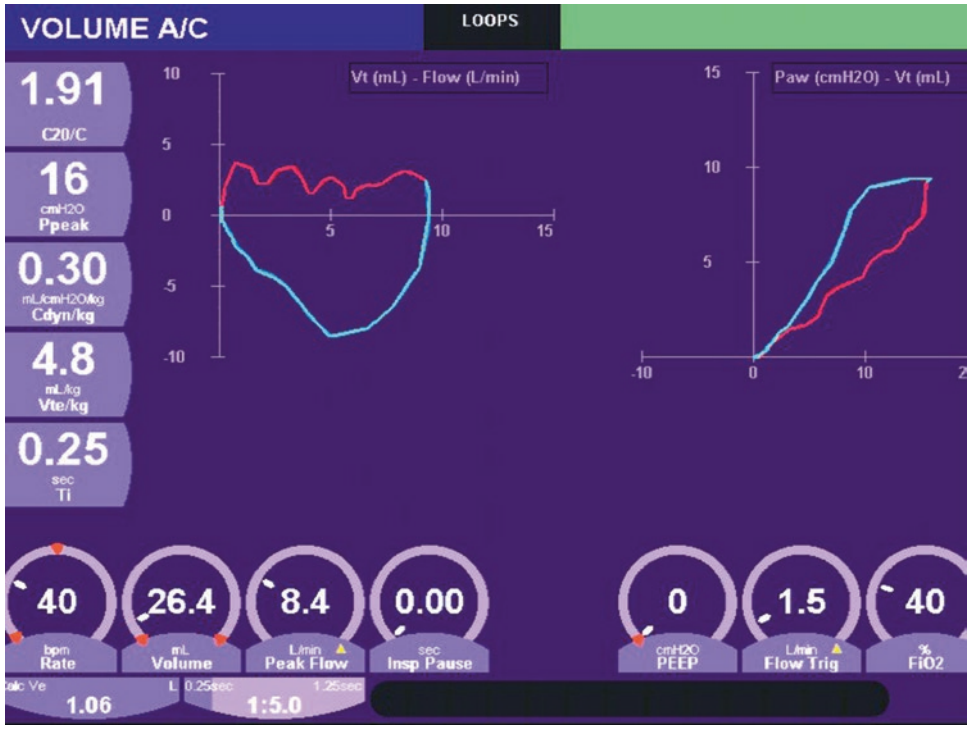


Fig. 22.19 Flow-volume loop on an infant in need of suctioning

2. Static compliance is the measurement based on static pressure:

$$C_{ST} = \frac{V_{Ti}}{P_{ST} - \text{PEEP}}$$

3. C_{20}/C is the ratio of compliance of the last 20 % of the *P-V* curve to the compliance of the entire curve. With overdistension this ratio will be less than 1.0.

E. Resistance is the relationship of pressure to flow. The pressure may be dynamic or static, and flow measurements are taken from various measurements.
1. Peak flow is the maximum flow on either inspiration or expiration.
2. Average flow is based on multiple point linear regression.
3. Mid-volume flow is based on the flow measured at a point of mid-volume delivery.
4. $R_{AW}\left(\text{cm}\,H_2O / L / s\right) = \dfrac{\text{PIP} - \text{PEEP}}{\text{Flow}}$

Suggested Reading

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